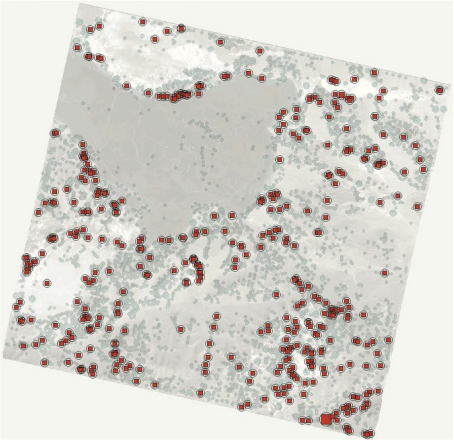
ALLOTROPE21 LABS

Allotrope

Anomaly detection, material identification and atmospheric correction in hyperspectral satellite imagery

Indradhanu, our own spectral foundation model, learns ordinary ground; what it cannot rebuild is flagged, then named man-made or natural with its closest material class.

AIGC Stage-III evaluation · PS-11
4 November 2026, 09:00



PRISMA, 14 Feb 2025: 405 of 5,544 anomaly objects called man-made.

0.189	449 / 909	86–94 %
pAUC, false alarms ≤1 %	objects in the top 1 %	man-made targets found
previous release 0.170	no model 377	at quarter-pixel fill; ACE 66–77 %

Problem and data

THE TASK

Find small, unusual objects in hyperspectral images with no target list given in advance, and say what material each is made of. At Stage II our model (700+ scenes) raised false alarms on urban scenes, land–water edges, cloud contours and snow; Stage III widened the data and moved the model's attention to the spectrum.

DATA PROCURED FOR STAGE III

pretraining	SpectralEarth, all of it: ≈3.3 TB, ≈11,636 EnMAP scenes, 539,755 patches
curated	1,300+ EnMAP, PRISMA and AVIRIS-NG scenes; India and borders; urban, forest, desert, hills, coast; 2–40 % cloud
L1 / L2 pairs	60 EnMAP and 30 PRISMA (L1, L2C, L2D) for atmospheric correction; 50+ public Tanager scenes

SENSORS

sensor	bands in the file	pixel	role
PRISMA	239 (185 used)	30 m	test bench; fine-tuning; namer
EnMAP	224	30 m	pretraining; fine-tuning; namer
AVIRIS-NG	372	varies	fine-tuning; namer; field targets
Tanager	426 (368 good)	30 m	end to end; never trained on

THE TEST BENCH

Four labelled PRISMA L2D scenes, **909 ground-truth objects** in three tiers (easy, medium, subtle). Every method is scored on the same valid pixels and the same object maps.

MEASURES

measure	what it means	chance
pAUC	area under the detection curve while false alarms stay ≤1 % of background, scaled so 1 is perfect	≈0.005
top 1 %	object found if any pixel lies within 1 px of the top 1 % of scores	10.2 % (93)
top 0.1 %	the same, ten times stricter	1.1 %
ROC-AUC	ranking over all thresholds	0.5
bootstrap	resample the 909 objects 2,000 times; 95 % interval of the difference	

Bench resolution: about ±0.005–0.01 pAUC.

Pipeline

Band filter
native bands; dead bands out; no-data filled

→

Segmentation
V2 head: land, water, cloud, shadow, snow

→

Scoring
V2 rebuilds each pixel hidden; Mahalanobis score

→

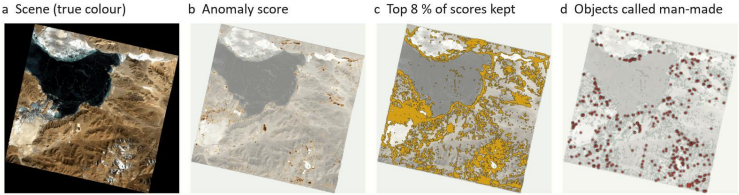
Threshold
analyst keeps the top scores; boolean mask

→

Spectral match
namer: man-made or natural, class; SAM, ACE

→

Export
GeoTIFF mask, shapefiles with material attributes



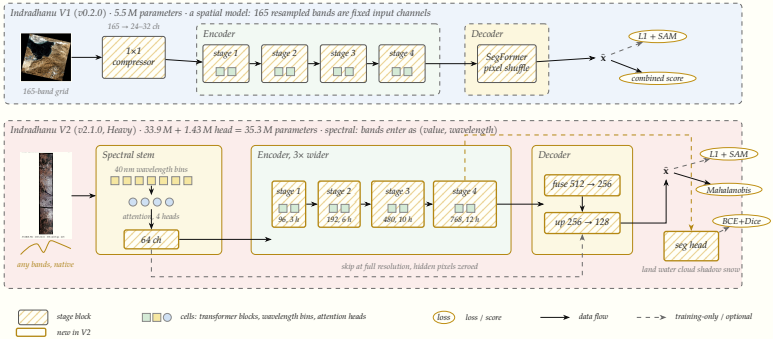
One PRISMA scene (14 Feb 2025, 1237×1202 px) through the chain: 30.3 % no-data; thick cloud (4.9 %) left out; the analyst keeps the top 8 %, 118,950 pixels; 5,544 objects named, 405 man-made (red).

TIME PER PRISMA SCENE

V2 Heavy, 16-thread CPU	454 s
previous release	274 s
residual Mahalanobis	3–4 s
spectral namer	8–9 s

The matcher's natural call removes natural anomalies after scoring, with a tunable confidence.

Spatial to spectral



V1, Stage II

A spatial SegFormer MAE on one resampled 165-band grid; a 1×1 compressor ties weights to band position: it knows "input 87", not 1,650 nm.

V2, Stage III

A wavelength-aware stem turns each pixel into tokens over 40 nm bins and attends over them: PRISMA's 185, EnMAP's 224, AVIRIS-NG's 372 bands read natively.

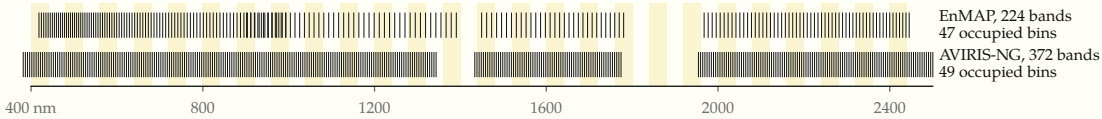
What made it a foundation model

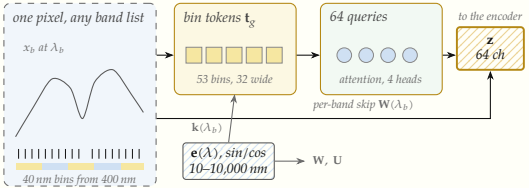
SpectralEarth pretraining (512,849 patches), band-subset and wavelength-jitter augmentation, 3× wider encoder, progressive decoder, segmentation head.

pAUC 0.165 (V1) → 0.170 (V2.0) → 0.189 (V2 Heavy) on the 909-object bench, thick cloud excluded.

Spectral stem

TWO REAL BAND LISTS, ONE TOKEN GRID: EVERY SENSOR'S BANDS FALL INTO THE SAME 40 NM BINS





THE MECHANISM, PER PIXEL

$$\mathbf{e}(\lambda) = [\sin \tfrac{2\pi\lambda}{P_i}, \cos \tfrac{2\pi\lambda}{P_i}]_{i=1}^{16}, \qquad P_i \text{ from 10 to } 10^4 \text{ nm}$$

$$\mathbf{t}_g = \langle x_b\, \mathbf{k}(\mathbf{e}(\lambda_b))\rangle_{b\in g} + \beta(\mathbf{e}(\bar{\lambda}_g)), \qquad g = 1,\dots,53 \text{ (40 nm)}$$

$$z_j = \mathrm{Attn}(\mathbf{q}_j;\, \{\mathbf{t}_g\}) + \langle x_b\, W_j(\mathbf{e}(\lambda_b))\rangle_{\text{all } b}, \qquad j = 1,\dots,64$$

$$\hat{x}(\lambda) = \mathbf{U}(\mathbf{e}(\lambda)) \cdot \hat{\mathbf{z}} + c(\mathbf{e}(\lambda)) \qquad \text{at any requested } \lambda$$

$$\langle f\rangle_B = \textstyle\sum_{b\in B} w_b f_b \,/\, \sum_{b\in B} w_b, \qquad w_b = \text{nm band } b \text{ covers}$$

x_b: band b's reflectance, normalised by one fixed curve over wavelength (not per scene). $\mathbf{k}, \mathbf{W}, \mathbf{U}, c$: small networks of $\mathbf{e}$, so no weight is tied to a band position. 64 learned queries attend over the occupied bins with 4 heads. A token *is* a wavelength range: a band at 1,655 nm reuses what was learned at 1,650 nm.

- **Band-list augmentation**: 80 % of batches see a random subset (≥40 bands), wavelengths shifted up to 3 nm. Before it, native PRISMA: ROC-AUC 0.205, resampled 0.725.
- 17,825 parameters; ≈6× faster than attention over every band.

Training

TWO STAGES ON ONE RENTED H100 (80 GB)

stage	data	model	epochs	settings, time
pretraining	SpectralEarth (EnMAP), 202 bands; 512,849 training patches, 30,000 per epoch	33.9 M	80	lr 3×10^{-4}, batch 32, bf16; ≈9 h
fine-tuning	EnMAP 210 scenes (14,542 tiles), PRISMA 223 (14,802), AVIRIS-NG 79 lines (22,377); mix 40/35/25 %	35.3 M (+1.43 M head)	10+8+6	lr $10^{-4}\rightarrow3\times10^{-5}$; ≈5.5 h
selection	four labelled PRISMA scenes	35.3 M	ep. 5 of 6	epochs 4 / 6: 0.184 / 0.187

OBJECTIVE, ON HIDDEN PIXELS M (65 % OF EACH PATCH)

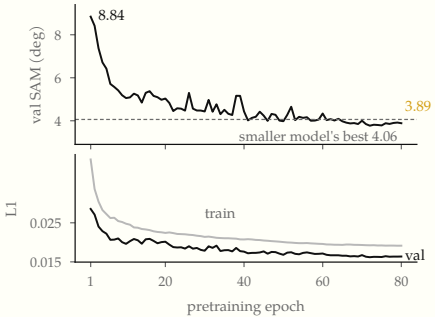
$$\mathcal{L} = \underbrace{\frac{1}{C|M|}\sum_{p\in M}\sum_{c}|\hat{x}_{c,p}-x_{c,p}|}_{L_1} + \lambda(t)\ \underbrace{\frac{1}{|M|}\sum_{p\in M}\angle(\mathbf{x}_p,\hat{\mathbf{x}}_p)}_{\text{spectral angle}} + 0.1\,\mathcal{L}_{\text{seg}}$$

$\lambda(t)$ rises 0 → 0.5 over the first ten epochs; $\mathcal{L}_{\text{seg}}$ = BCE + Dice, fine-tuning only, shadow and snow ×3.

FEEDING THE GPU

150-patch archives (≈1 GB, 3,419 for training) read straight from storage gave 18 patches/s; 200 at a time are copied to local disk and swapped every five epochs: 81 patches/s, 18.4 GB peak, 3,200 archives used. Scenes within 30 km of the bench are kept out.

PRETRAINING, VALIDATION

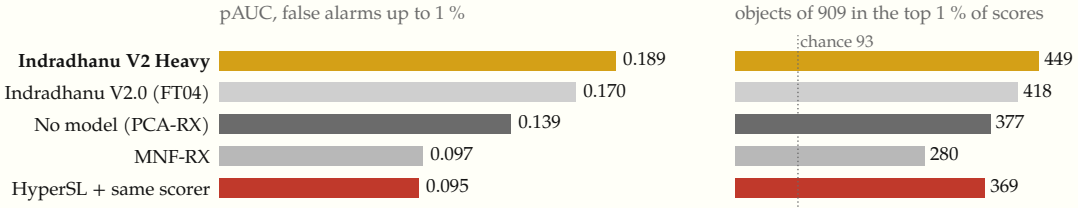


Validation spectral angle 8.84° → 3.89°, below the smaller model's best 4.06°; epoch 72 seeds fine-tuning.

Epoch 5 was picked on the bench, so its bench score is slightly optimistic; its neighbours sit within 0.005.

Detection results

909 labelled objects, four PRISMA scenes, thick cloud excluded; water stays in (vessels are targets)



The score: residual Mahalanobis

$$\mathbf{r}_p = \mathbf{x}_p - \hat{\mathbf{x}}_p, \qquad \mathbf{z}_p = \Lambda_d^{-1/2}\mathbf{U}_d^{\top}(\mathbf{r}_p - \mu_r),\ d = 30$$

$$s(p) = \left\| \mathbf{z}_p - \tfrac{1}{|\mathcal{R}_p|} \textstyle\sum_{q \in \mathcal{R}_p} \mathbf{z}_q \right\|^2$$

Whiten the residual in its 30 strongest directions; compare with a ring between 21×21 and 61×61. Direction, not size: mean ROC-AUC **0.662 → 0.854**, same model.

Paired bootstrap against FT04

epoch 5	**+0.0196** [+0.0072, +0.0316]
epoch 4	+0.0148 [+0.0029, +0.0259]
epoch 6	+0.0179 [+0.0055, +0.0292]
no model − V2	−0.047 [−0.060, −0.033]

Top 1 % by tier: easy 58.2 → **62.2**, medium 39.3 → **43.9**, subtle 40.3 → **42.0** %. At top 0.1 % the two tie (+0.1 pts).

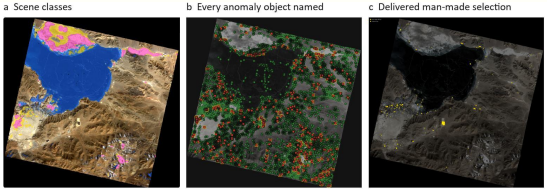
Per scene, V2 Heavy

	pAUC	ROC-AUC
Set-1	0.2617	0.936
Set-2	0.1874	0.874
Set-3	0.0861	0.781
Set-4	0.2217	0.888

The delivered chain reproduces all four to four decimals. Leaving out thick cloud only: 0.175 → 0.189.

Public benchmark HAD100: appendix A6.

Detection examples



PRISMA. (a) V2 classes: water blue, cloud pink (left out), haze yellow. (b) 5,544 objects, man-made red, natural green. (c) A delivered selection.

THE SAME CHAIN ON THREE SENSORS

	Tanager	PRISMA	AVIRIS-NG
anomaly objects	1,103	5,544	2,375
man-made, $P > 0.5$	272	405	1,290
man-made, $P > 0.25$	585	2,263	1,716
oil/fuel or plastic	247	141	936

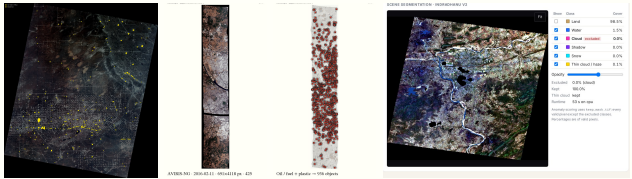
Tanager was never in training; GHOSt (Golmud, 8 m) ran end to end on evaluation day.

THE 405 MAN-MADE OBJECTS

concrete/asphalt	244	metal	16
oil/fuel	116	textile	3
plastic	25	wood/paper	1

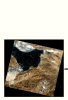
SEGMENTATION HEAD, VALIDATION IoU

water 0.825, cloud 0.749. Its thin-cloud class also fires on haze (75.4 % of Set-4, 209 of 230 targets), so only thick cloud is left out.



Tanager, man-made | AVIRIS-NG | oil + plastic | EnMAP, classes

Atmospheric correction



L1 radiance or TOA, any file

Radiometric

Read native or generic; level detected, not assumed

To radiance gain, offset, scale per sensor

Atmospheric (ISOFIT 4.1.5)

Look-up table 6S + sRTMnet; AOT, H_2O, elevation, geometry

Optimal estimation on superpixels; surface prior

Every pixel analytical or empirical line, GPU

Surface

Post outliers, terrain, adjacency, snow

Harmonise towards the vendor L2, per sensor

L2 surface reflectance GeoTIFF + BSQ, wavelengths

Copernicus DEM

vendor L2 (fit only)

RADIANCE MODEL AND OPTIMAL ESTIMATION

$$\rho_{\mathrm{TOA}} = \frac{\pi d^2 L}{E_0 \cos\theta_s} = \rho_{\mathrm{path}} + \frac{T^{\downarrow} T^{\uparrow} \rho_s}{1 - S\rho_s}$$

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \; \|\mathbf{y} - F(\mathbf{x})\|^2_{\mathbf{S}_\epsilon^{-1}} + \|\mathbf{x} - \mathbf{x}_a\|^2_{\mathbf{S}_a^{-1}}$$

State $\mathbf{x}$: surface reflectance $\rho_s(\lambda)$, aerosol optical thickness, water vapour. F: the look-up-table forward model; $\mathbf{x}_a$, $\mathbf{S}_a$: surface prior; $\mathbf{S}_\epsilon$: instrument noise. Adjacency: $\rho = \rho_{\mathrm{ret}} + q(\rho_{\mathrm{ret}} - \rho_{\mathrm{bg}})$, 1 km kernel. Inverts within 350–1360, 1410–1800, 1970–2500 nm. Superpixels of 50 for satellites; 400 with an empirical line for 4–16 Mpix AVIRIS-NG lines.

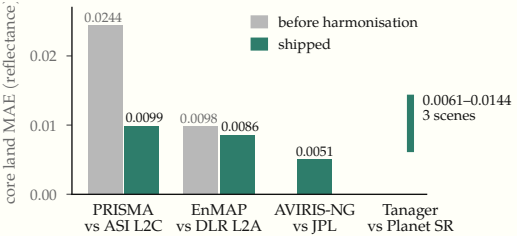
PER-SENSOR RULES

sensor	to radiance	aerosol	terrain, adjacency	bands kept
PRISMA	DN / scale − offset; VNIR/SWIR split 950 nm	NASA POWER climatology ±0.15	off	206
EnMAP	DN × gain + offset	free, seasonal limits	on	218
AVIRIS-NG	units only	0.10–0.14, as JPL	off	380
Tanager	units only	free, seasonal limits	on	382

Terrain only where the vendor does it: SWIR r 0.947 → 0.980 on mountains. Snow by NDSI (precision 0.91–1.00). Reflectance > 2.5 kept and flagged: flares and fires are anomalies.

Correction accuracy

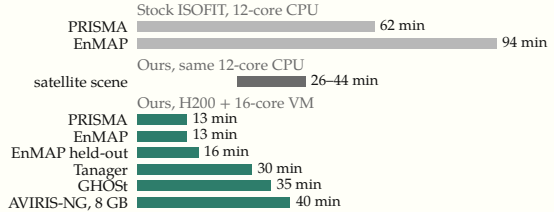
HELD-OUT SCENES AGAINST THE VENDOR L2, 460–2450 NM CORE



Two good L2 products differ by 0.01–0.02. A per-sensor model of the retrieved atmosphere (AOT, H_2O, sun zenith, elevation) maps ours to the vendor's; fitted on 15–27 scenes, scored once on 5 held-out. PRISMA: all five improved. EnMAP: blue bands only, 460 nm bias +0.023 → +0.001.

EnMAP's blue range was chosen after the held-out scores were seen. Aim: one reflectance domain, so one anomaly model serves every sensor.

RUN TIME PER SCENE



HOW

exact	6S aerosol-only call removed (1.9×), scattering cached over water vapour (5×); GPU per-pixel solve 3.2×, difference $\leq 6\times10^{-8}$
measured	EnMAP prior on true band centres 53 → 30 min; AVIRIS-NG superpixels 400: window 40.8 → 7.5 min, error 0.0056 → 0.0057; full 4 Mpix line 373 → 41 min
robust	input level detected by O_2/H_2O absorption tests, 28/28 right; ambiguity stops the run and names the fix

Material identification

TWO MATCHERS ON EVERY DETECTED PIXEL

USGS splib07 + ECOSTRESS, resampled to the sensor: seven man-made classes (oil/fuel, plastic, textile, wood/paper, paint, metal, concrete/asphalt) and a natural shelf; 378 / 338 / 380 spectra (EnMAP / PRISMA / AVIRIS-NG).

SAM: smallest angle $\theta_j = \arccos \frac{\mathbf{x}\cdot\mathbf{s}_j}{\|\mathbf{x}\|\|\mathbf{s}_j\|}$; no training, but blind to the ground filling the rest of the pixel.

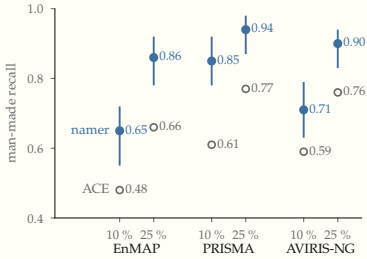
Spectral namer (347,685 parameters) on **V2's spectral stem**: one tower reads the pixel and its ring mean (3–7 px away), the other the library.

$$s_j = \cos(\mathbf{z}, \mathbf{e}_j)/\tau, \quad S_c = \log \sum_{j \in c} e^{s_j}$$

$$P(\text{man-made}) = \sigma\left(\mathrm{LSE}_{c \neq \text{nat}}\, S_c - S_{\text{nat}} + b \right)$$

Level 1: share of the match on man-made spectra; level 2: best class. Trained on library spectra planted at 5–100 % fill in real EnMAP, PRISMA, AVIRIS-NG pixels, 20 % of each class held out. New spectra need no retraining.

LEVEL 1 AT ACE'S FALSE-ALARM RATE



Quarter-pixel fill: 86–94 % of man-made targets found; ACE 66–77 %.

LEVEL 2, CLASS CORRECT

fill 10 %	namer	ACE
EnMAP	**0.59** [0.48, 0.70]	0.24
PRISMA	**0.69** [0.53, 0.81]	0.41
AVIRIS-NG	**0.65** [0.53, 0.76]	0.38
fill 25 %		
EnMAP	**0.68** [0.55, 0.79]	0.47
PRISMA	0.70 [0.54, 0.82]	0.60
AVIRIS-NG	0.70 [0.57, 0.82]	0.58

Planted sub-pixel targets in PRISMA scenes, 10 % fill, 8 classes: namer **0.53**; best training-free 0.44; HyperSL-based 0.27–0.30.

Field targets

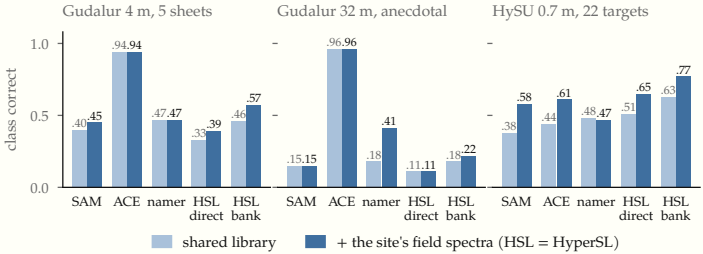
REAL TARGETS LAID OUT IN THE FIELD

HySU (DLR, airborne, 0.7 m): bitumen, red-painted metal, blue, red and green synthetic fabric, squares 0.25–3 m; 135 bands, 417–903 nm, a range the namer never trained on. 22 targets, 236 target pixels, 3,000 background.

Gudalur (AVIRIS-NG, Nilgiris 2018, 4 m): five 16×16 m sheets (green, red, yellow, black nylon; white cotton) with field spectra; 80 pixels, 353 bands. Also coarsened to 32 m at four offsets: each sheet ≤ a quarter pixel, 27 pixels (anecdotal).

Namer here has textile held out, so textile is unseen; nylon may count as plastic or textile, bitumen as concrete or oil.

LEVEL 2: CLASS CORRECT, SHARED LIBRARY AND WITH THE SITE'S OWN SPECTRA



level 1: AUC of the man-made call	Gudalur 4 m	Gudalur 32 m	HySU
SAM	0.69 / 0.70	0.39 / 0.39	0.65 / 0.73
ACE	0.58 / **0.89**	0.58 / **0.86**	0.63 / **0.81**
spectral namer	0.15 / 0.15	0.17 / 0.18	0.58 / 0.61
HyperSL, mixture bank	0.51 / 0.59	0.15 / 0.19	0.57 / 0.80

ACE names the Gudalur sheets best (5 of 5 at 4 m, 18 of 19 at 32 m). The namer's man-made call is *inverted* on the sheets: a 16 m sheet is not a pixel unlike its ring. On HySU the HyperSL mixture bank names the most (16 of 22; 18 with field spectra). Small sets: examples, not rates.

Against HyperSL

A PUBLIC SPECTRAL FOUNDATION MODEL, RUN UNMODIFIED

Per-pixel masked autoencoder, 9.2 M parameters, EnMAP and DESIS reflectance; hides 95 % of a pixel's bands and rebuilds them from the same pixel.

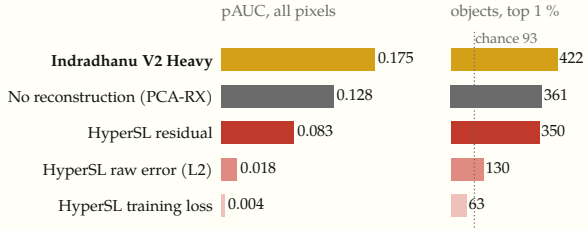
1 Self-reconstruction. An unusual pixel rebuilds as well as an ordinary one: its own loss as a score finds 63 objects, fewer than random (93). Indradhanu hides whole pixels.

2 Per-scene min-max. $x'_b = (x_b - m_b)/(M_b - m_b)$ from the scene: $(0.20, 0.50)$ becomes $(0.5, 1.0)$ in one scene, $(0.43, 0.75)$ in another; an angle moves $15.1° \rightarrow 36.9°$ with no change in reflectance. A library spectrum has no scene: 41 % of ECOSTRESS values clipped.

3 Band order, not wavelength. List neighbours straddle water-vapour gaps and differ per sensor. Indradhanu bins by wavelength.

4 Cloud-free pretraining. Cirrus recall 28.9 % from its features against 67.9 % from raw bands.

SAME BENCH, SAME SCORER, ALL PIXELS



paired difference [95 %]	pAUC	objects (pts)
HyperSL − no model	−0.045 [−0.059, −0.032]	−1.2 [−4.0, +1.5]
HyperSL − V2 Heavy	−0.092 [−0.106, −0.078]	−7.9 [−10.8, −5.2]

unmixing RMSE, true endmembers	DC1 / DC2 / DC3	EGU	Jasper	Urban
reflectance, best physical	0.000 / 0.000 / 0.000	0.018	0.050	0.059
HyperSL internal space	0.253 / 0.113 / 0.261	0.142	0.106	0.170

Not adopted: it lowers detection below no model, bends mixing (0.4–0.7 of the path length), and its matching trails plain reflectance.

Deliverables and limits

Per scene and selection

- Boolean anomaly mask, GeoTIFF, 0/1 on the scene's own UTM grid; **no heatmap**.
- Pixel and object shapefiles and a table, latitude and longitude.
- Every pixel and object: namer call, class and confidence, ACE family match, best SAM match and its angle.
- L2 surface reflectance from L1, GeoTIFF and BSQ with wavelengths.

On the evaluation VM

	scoring	L1 → L2
time	<30 s	16 min
CPU, 16 cores	≈10 %	≈90 % peak
RAM, 64 GB	≈24 %	≈10 %
GPU, H200	≈3 %	≈4 %

Evaluation scene: GHOST, Golmud airport

GeoTIFF input; airstrip, a smaller field and buildings on bare soil; visible striping. Two scorings: Heavy + RX hybrid, and Heavy alone. Natural calls removed: ≈8,000 → ≈3,000 and ≈5,500 → ≈3,700 pixels submitted.

man-made class	sub. 1	sub. 2	namer's match	SAM's match
plastic	39.6	30.3	fiberglass	cerium oxide
metal	24.2	21.8	coated steel girder	calcite
concrete/asphalt	23.1	23.4	construction concrete	lead white
oil/fuel	7.7	16.8	oiled marsh plant	buckbrush
textile	2.2	3.7	terrycloth fabric	grass
wood/paper	2.2	3.7	particle board	grass
paint	1.1	0.3	olive green paint	grass

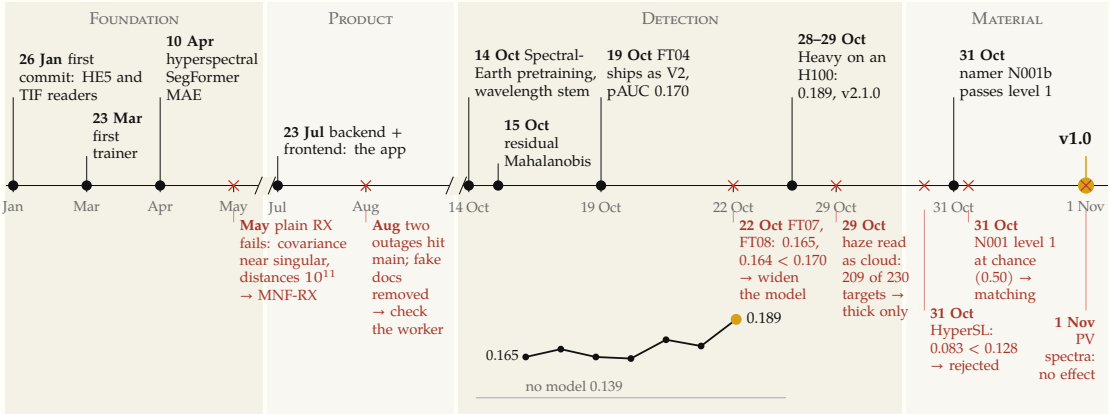
Share of man-made pixels, %. SAM's nearest spectra are minerals and plants.

Limits

- **Large targets.** The namer compares a pixel with its ring; roads, roofs and big sheets defeat it. SAM and ACE guide there.
- **Open set.** Unseen textile: 46–47 % still called man-made, then a confident wrong class; abstention flags 1–4 %. In-scene solar panels changed nothing (0.248 → 0.248).
- **Small sets.** 5 sheets at Gudalur, 22 targets at HySU; four PRISMA scenes, ±0.005–0.01 pAUC.
- **Threshold** is an analyst choice; the evaluation scenes have no ground truth.
- **Cost.** Heavy 454 s per PRISMA scene on CPU, FT04 274 s.

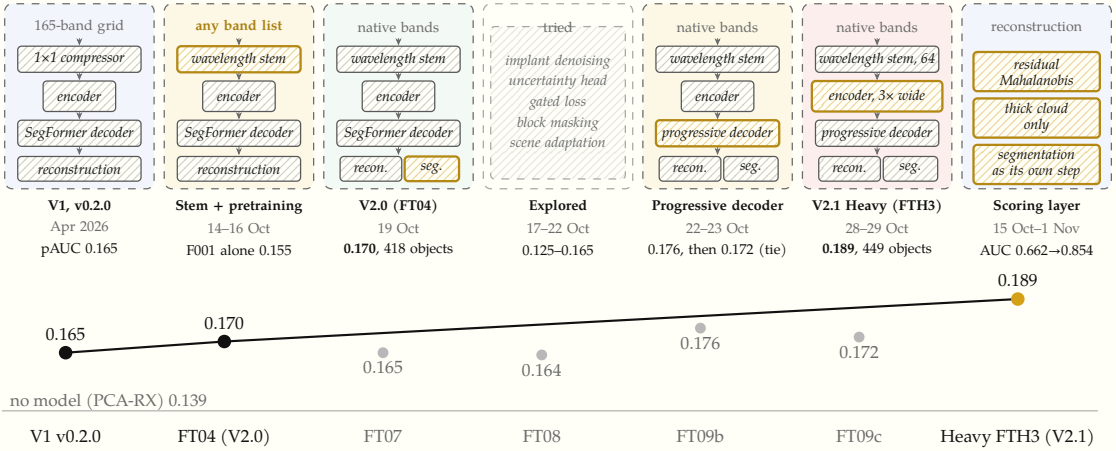
Next: field spectra for camouflage and coatings, a glass and solar-panel class, a labelled bench beyond PRISMA.

Journey



From a spatial pipeline to a spectral foundation model: **any sensor in, materials out.**

A1 Model generations



Shipped generations in ink, experiments that did not beat the shipped model in grey; each column adds the part outlined in ochre.

Also tried, not shipped: uncertainty head 0.125 / 0.163; per-scene adaptation 0.160. Pretrained only: 0.155 (FT04 size), 0.166 (Heavy).

A2 Namer runs

run	change	result
N001	linear level-1 head	level 1 at chance, 0.49–0.52 (0.50 at full fill, where ACE gets 0.87)
N001b	level 1 by matching	passes on all three sensors, +0.20 / +0.17 / +0.14 over ACE; **shipped**
N002	textile held out	open set: 46–47 % of textile still called man-made
N003	mixed spectra in a class, warped wavelengths, dropout	matched N001b on EnMAP, no gain; N001b ships
PV	50 in-scene solar-panel spectra	no effect, 0.248 → 0.248

N003 AGAINST THE SHIPPED N001B, ENMAP, EPOCH 40

measure	N001b (shipped)	N003
level 1 accuracy, fill 10 %	0.679	0.677
level 2 class accuracy	0.587	0.588
man-made recall at ACE's false-alarm rate	0.647	0.639

PLANTED PIXELS FROM REAL TILES OF THREE SENSORS

$$\mathbf{x} = \mathbf{g} \odot (a\,\mathbf{t} + (1-a)\,\mathbf{b}) + \mathbf{n}, \quad a \sim \text{log-uniform}[0.05, 1]$$

t a library spectrum (50 % man-made, 25 % natural, 25 % none), **b** the real background, **g** a smooth gain within ±10 %, **n** the tile's own noise.

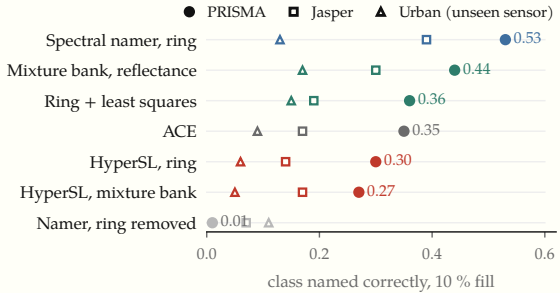
$$\mathcal{L} = \text{BCE}(\ell, \text{M}) + \text{CE}(S_{0..6}, c) + 0.5\,|\hat{a} - a|$$

parameters	347,685, of which the shared stem 17,825
optimiser	AdamW 3×10^{-4}, stem at 0.3×
run	40 epochs × 500 steps on one H100
intervals	500 resamples of 221–254 spectra per sensor

About 34 unseen man-made spectra per sensor: differences much below 0.12 are not resolved.

A3 HyperSL detail

PLANTED TARGETS, 10 % FILL, 8 CHOICES, TARGET'S OWN SPECTRUM REMOVED



The background estimate matters most: ring of neighbours > scene mean > nothing. Without its ring the namer collapses (0.01): its call is, in effect, "this pixel differs from its neighbours". About 40 % of targets cannot be named even when pure; that caps every method near 0.6.

PER-SCENE MIN-MAX, WORKED

Scene 1: $p_1 = (0.10, 0.30)$, $p_2 = (0.20, 0.50)$, $p_3 = (0.30, 0.40)$ give $p_2' = (0.5, 1.0)$. The same p_2 in scene 2 gives $(0.43, 0.75)$. One bright anomaly moves every other pixel's input. Angle p_2, p_3: 15.1° before, 36.9° after.

THE DECISION

need	HyperSL	Indradhanu
same spectrum, same input	min-max per scene	fixed curve over wavelength
library spectra have an input	none	same stem
band lists with gaps	groups by band position	40 nm wavelength bins
a pixel never rebuilds itself	rebuilds from its own bands	hides whole pixels
cloud in pretraining	cloud-free	cloud and haze seen

HyperSL ran on a 149-band PRISMA axis with its default mask ratio; nothing tuned on the bench. Subtle tier suffers most: 29.7 % found against 35.0 % with no model.

A4 Evaluation method

PARTIAL AUC, THE HEADLINE MEASURE

$$\mathrm{pAUC}_m = \frac{1}{m}\int_0^m \mathrm{TPR}(f)\,df = \frac{1}{P}\sum_p \frac{\max(0,\, m - \mathrm{FPR}(p))}{m}$$

$m = 0.01$; $\mathrm{FPR}(p)$ is the share of background pixels scoring at or above target pixel p.

PAIRED OBJECT BOOTSTRAP

- Resample the 909 objects with replacement, 2,000 draws; score both methods on each draw; report the 95 % interval of $B - A$.
- Objects, not pixels, are the independent unit; pairing removes shared object difficulty.
- It does not cover checkpoint-to-checkpoint variation, so three neighbouring checkpoints were checked.

ALL MEASURES; EXCL. = THICK CLOUD LEFT OUT

method	pAUC all	pAUC excl.	top 1 % all	top 1 % excl.	top 0.1 % excl.	ROC excl.
V2 Heavy	0.175	**0.189**	422	**449**	16.0 %	0.869
V2.0 (FT04)*	0.162	0.170	415	418	15.8 %	0.852
no model (PCA-RX)	0.128	0.139	361	377	8.5 %	0.842
MNF-RX	0.075	0.097	249	280	5.2 %	0.753
HyperSL residual	0.083	0.095	350	369	6.7 %	0.761
HyperSL own error	0.018	0.018	130	137	4.3 %	0.628
random			93	93	1.1 %	0.5

* FT04 with its own cloud and thin-cloud mask.

WHY ONLY THICK CLOUD

Heavy's thin-cloud class fires on haze: 75.4 % of Set-4's pixels and 209 of its 230 targets. Cloud is out of scope; water is in, because vessels are targets.

A5 Deployment

air-gapped evaluation server

- One H200 (140 GB) with a 16-core CPU and 64 GB RAM; no network.
- The product (web API, worker, frontend, database) and the atmospheric-correction container arrive as offline archives with checksums; IT scans each image.
- Update and rollback replace containers and never touch data volumes or settings; rehearsed on a simulated copy of the server.
- Atmospheric correction runs with networking disabled, as non-root, on GPU or CPU only. DEM tiles (Copernicus 30 m South Asia, 90 m worldwide) are staged locally.

atmospheric-correction container

core	ISOFIT 4.1.5, sRTMnet emulator, 6S compiled for x86-64-v3/v4
tests	81 passed, 54 % coverage
size	22 GB image, 8.98 GB compressed

worker cost, one PRISMA scene, 185 bands (s)

	reconstruct	segment	score	total
Heavy, 2 threads	1629	185	4	1819
Heavy, 16 threads	402	47	4	454
FT04, 16 threads	236	32		274

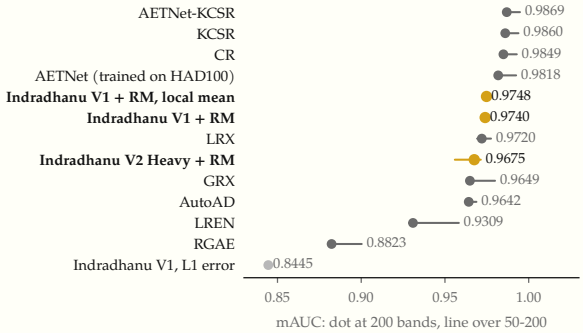
superpixel size against accuracy, AVIRIS-NG window vs JPL

superpixels, per-pixel step	time	land MAE	median angle
50, analytical	40.8 min	0.0056	2.15°
200, analytical	17.8 min	0.0056	2.17°
200, empirical line	11.3 min	0.0056	2.17°
400, empirical line	7.5 min	0.0057	2.19°

Before the speed work: up to 75+ min for PRISMA and EnMAP and nearly 6 h for a full AVIRIS-NG scene; 12–14 GB scenes now fit the sub-2-hour window.

A6 HAD100

PUBLIC BENCHMARK, OFFICIAL PROTOCOL, AGAINST LI ET AL. 2023



At 200 bands residual Mahalanobis sits 0.013 (V1) and 0.019 (V2 Heavy) below the best, above GRX and the deep reconstruction baselines.

On 64×64 crops plain RX already scores 0.96–0.98, so HAD100 ranks methods differently from our PRISMA bench.

PROTOCOL

100 crops of 64×64 from 94 AVIRIS-NG tiles (3,567 anomaly pixels, 460 objects), L1B radiance, 276 "good" bands; papers then take the first 50 / 100 / 200. Measure: mAUC, the mean per-crop ROC-AUC. Our GRX reproduces the paper's GRX to four decimals.

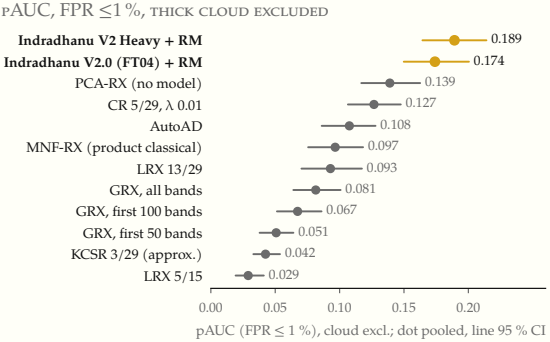
OUR RUNS, MAUC

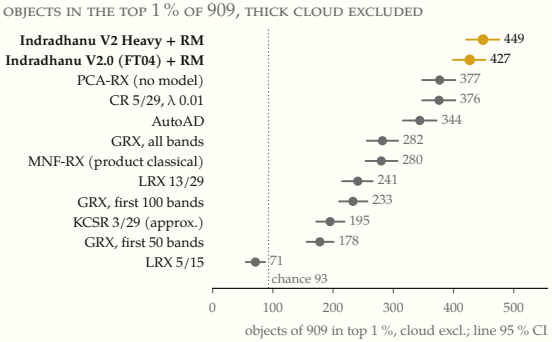
	50 b	100 b	200 b	276 b / full	no Kakinada
GRX	0.9799	0.9714	0.9649	0.9634	0.9611
PCA30-RX	0.9797	0.9603	0.9636	0.9648	0.9627
Indradhanu V2 Heavy + RM	0.9714	0.9559	0.9675	0.9703	0.9684
local mean 5/15	0.9715	0.9565	0.9688	0.9714	0.9696
Indradhanu V1 + RM				0.9740	0.9724
local mean 5/15				0.9748	0.9732

V1 reads only its 165-band grid, so one value fills every column. "No Kakinada": 94 crops, at full bands, after removing 6 from a line flown 1–2 km from a training swath.

Reading. V2 Heavy ties V1 (0.970 vs 0.974 at full bands; 47/52 crops, $p = 0.5$) on an airborne radiance benchmark outside its training domain (2–8 m, US/Swiss), while reading every band count natively; V1 needs its fixed 165-band grid. V2 beats GRX and PCA30-RX at 200 and 276 bands, not at 50. The leaders are local collaborative-representation detectors; AETNet trains on HAD100's own normal tiles.

A7 HAD100 methods on our bench





None of the HAD100 methods comes close: the best, CR, reaches 0.127 against 0.189 (-0.063 [-0.090, -0.035]) *and ties the no-model PCA-RX on objects* (*376 vs 377; V2 Heavy 449*).

The HAD100 ranking (KCSR > CR > LRX > GRX) does not carry over to 30 m PRISMA. Windows are HAD100's settings; CR's λ was picked on the bench (optimistic); CR and KCSR run on a 30-dimensional projection keeping 99.97 % of the energy; KCSR approximated. AutoAD, the deep per-image baseline (tiled 2×2): 0.108 and 344 objects, -0.082 [-0.107, -0.057] against V2 Heavy. Dot pooled over four scenes, line 95 % paired bootstrap.